NumPy Practice Questions

Beginner to Advanced • Python Practice Worksheet

Instructions

1. Solve the questions in Python using NumPy.
2. Start with Level 1 and move to the next level when you are comfortable.
3. Write and run each program yourself before checking a solution.

Required import

```
import numpy as np
```

Practice Questions

Level 1 — Basic NumPy

Array creation, indexing, shape, and data types

1. Import NumPy and create a 1D array containing numbers from 1 to 10.

2. Create an array of 5 integers and print its data type.

3. Create the array [10, 20, 30, 40, 50] and print the third element.

4. Create the array [5, 10, 15, 20, 25] and print the first and last elements.

5. Create an array of numbers from 0 to 9 using `arange()`.

6. Print the total number of elements, dimensions, and shape of an array.

7. Create a 3×3 array containing numbers from 1 to 9.

8. Create one array containing 5 zeros and another containing 5 ones.

Level 2 — Array Operations

Arithmetic, statistics, reshaping, and filtering

1. Add 5 to every element of the array [10, 20, 30, 40].

2. Add the two arrays [1, 2, 3] and [4, 5, 6].

3. Find the sum and mean of [10, 20, 30, 40, 50].

4. Find the maximum and minimum of [12, 45, 7, 89, 23].

5. Multiply every element of [2, 4, 6, 8, 10] by 3.

6. Reshape [1, 2, 3, 4, 5, 6] into a 2×3 matrix.

7. Reverse the array [10, 20, 30, 40, 50].

8. Filter the even numbers from [1, 2, 3, 4, 5, 6].

Level 3 — Intermediate

Conditions, matrices, and array manipulation

1. Find the numbers divisible by 3 in the range from 1 to 20.
2. Remove duplicate values from [1, 2, 2, 3, 4, 4, 5].
3. Sort the array [10, 20, 30, 40, 50] after rearranging its elements.
4. Perform addition and subtraction on two 2×2 matrices.
5. Perform matrix multiplication on two matrices.
6. Generate 10 equally spaced numbers from 5 to 50 using `linspace()`.
7. Replace 15 with 100 in the array [5, 10, 15, 20, 25].
8. Calculate the sum of each row and each column in a 3×3 matrix.

Level 4 — Advanced Practice

Random data, matrix operations, and real-world filtering

1. Generate 10 random integers and find their mean, median, and standard deviation.
2. Create a 4×4 identity matrix.
3. Find the transpose of a 3×3 matrix.
4. Find the maximum element in a matrix and its index.
5. Find the common elements in two arrays.
6. Convert an array to 1D using `flatten()` and `ravel()`; explain the difference.
7. Create a 5×5 matrix and print its diagonal values.
8. From the marks array [45, 78, 90, 32, 67, 88], filter marks greater than or equal to 60.

Quick Revision: Common NumPy Functions

Function	Use
np.array()	Create an array
np.arange()	Generate a range of numbers
np.zeros()	Create an array filled with zeros
np.ones()	Create an array filled with ones
np.reshape()	Change an array's shape
np.sum()	Calculate the sum
np.mean()	Calculate the average
np.max() / np.min()	Find the maximum / minimum
np.sort()	Sort values
np.unique()	Get unique values
np.where()	Select values based on a condition
np.random.randint()	Generate random integers

Goal: Complete all 32 questions to strengthen your NumPy fundamentals and array-manipulation skills.